MODULE 21 · DOMAIN 2 - PROCESS

Estimating effort and duration

Analogous, parametric, bottom-up and three-point estimates; story points and planning poker; ranges, history, and holding your ground under pressure.

DOMAIN II - Process EXAM WEIGHT 41% MODULE 21 of 49

admin@sthithapragna.com

An estimate is a range with a confidence, not a promise.

Effort or duration?

EFFORT

Work required

- Measured in person-hours or person-days
- 40 hours of design work
- Does not depend on calendar
- Drives cost

DURATION

Time it takes

- Measured in working days or weeks
- Depends on who is available, and how much
- One designer at half time: 2 weeks
- Drives the schedule

THE TIEBREAKER

Forty hours of effort is a week for one full-time person, and two weeks for one person at half time.

Estimating techniques

TECHNIQUE	HOW IT WORKS	BEST WHEN
Analogous	Use a similar past project as the basis	Early, quick, little detail known
Parametric	Multiply a unit rate by a quantity	Reliable rates exist: cost per square metre
Bottom-up	Estimate each work package, then add up	Detail is known; accuracy matters
Three-point	Optimistic, most likely, pessimistic	Uncertainty is significant
Expert judgement	Ask people with relevant experience	Almost always, alongside the others

ON THE EXAM

Analogous is fast and rough. Bottom-up is slow and accurate. Parametric is as good as its unit rate.

Analogous and parametric, side by side

- **Analogous**: last year's regional office fit-out cost $400,000. This one is similar, so about $400,000.
- **Parametric**: fit-outs cost about $1,600 per square metre. This office is 300 square metres: about $480,000.
- Parametric is usually more accurate, if the rate is reliable and the work scales with the quantity.
- Analogous is only as good as the similarity. Adjust for known differences.

> ON THE EXAM
>
> Both use history. Analogous uses one whole project; parametric uses a rate derived from many.

Three-point estimates

- **Triangular** average: (Optimistic + Most likely + Pessimistic) / 3.
- **Beta, or PERT**: (Optimistic + 4 x Most likely + Pessimistic) / 6.
- O = 4 days, M = 6, P = 14. Triangular: 24 / 3 = **8**. PERT: (4 + 24 + 14) / 6 = **7**.
- **Standard deviation**, PERT: (P - O) / 6. Here, 10 / 6 = about **1.7 days**.

> ON THE EXAM
>
> PERT weights the most likely value four times. Read which formula the question wants.

Turning a standard deviation into a range

- Plus or minus **1** standard deviation: about **68%** confidence.
- Plus or minus **2**: about **95%**. Plus or minus **3**: about **99.7%**.
- PERT estimate 7 days, standard deviation 1.7: 95% confident of 3.6 to 10.4 days.
- A wider spread between optimistic and pessimistic means more uncertainty, and a wider range.

> ON THE EXAM
>
> Communicate estimates as ranges with confidence. A single number hides the uncertainty.

Estimates narrow as knowledge grows

ESTIMATE	TYPICAL RANGE	WHEN
Rough order of magnitude	-25% to +75%	Initiation, before detail exists
Budget or preliminary	Narrower, as scope firms up	During planning
Definitive	-5% to +10%	Detailed scope and work packages known

ON THE EXAM

The 'cone of uncertainty': early estimates are wide. Refine them as the project learns.

Who estimates, and how honestly

- The **people who will do the work** estimate it. They know what it involves.
- Estimate each item **without hidden padding**. Uncertainty is handled openly, in reserves.
- Record the **basis of estimates**: assumptions, method and confidence.
- Padding hidden in every task gets used up anyway: work expands to fill the time available.

> ON THE EXAM
>
> Honest estimates plus explicit reserves beat padded estimates. Module 26 covers reserves.

Story points

- A **relative** measure of size: effort, complexity and uncertainty combined.
- Often on a Fibonacci-like scale: 1, 2, 3, 5, 8, 13, 20. The gaps grow because uncertainty grows.
- Compare to a reference story: 'this is about twice our login story'.
- Not hours, and not comparable between teams. Each team calibrates its own.

> ON THE EXAM
>
> Story points measure size relative to other stories, for one team. They are not a unit of time.

Planning poker

01

Explain

The product owner describes the story; the team asks questions.

02

Vote privately

Each developer picks a card, all shown at once.

03

Discuss outliers

The highest and lowest explain their reasoning.

04

Re-vote

Repeat until estimates converge.

ON THE EXAM

Outliers are the point. The person who sees risk others missed usually holds the highest card.

Other agile estimating techniques

TECHNIQUE	HOW IT WORKS
T-shirt sizing	Extra small to extra large; quick and rough, good for epics
Affinity estimating	Place many items in size groups on a wall, fast
Ideal days	How long it would take with no interruptions
Throughput forecasting	Count items finished per period; forecast from history

ON THE EXAM

Coarse techniques for large or early items; finer ones near the top of the backlog.

Forecasting from velocity

- **Velocity**: story points completed per iteration, measured, not planned.
- Iterations remaining = points remaining / average velocity.
- 240 points left; velocity has ranged 30 to 40. Forecast: **6 to 8** iterations.
- Forecast as a range, using the team's own recent history, and update it every iteration.

> ON THE EXAM
>
> Use the team's measured velocity, not a hoped-for one, and forecast a range, not a date.

Benchmarks and historical data

- The outline asks you to **utilise benchmarks and historical data**.
- Your own past projects, the lessons learned repository, and industry benchmarks.
- Compare past estimates with actuals, and **calibrate**: if similar work ran 40% over, find out why.
- Ignoring history in favour of optimism is the most common estimating error of all.

> ON THE EXAM
>
> History is the best predictor you have. Use it, calibrate against it, and explain differences.

When someone wants a smaller number

- Cutting an estimate because a sponsor dislikes it does not change the work. It hides the risk.
- Explain the basis of the estimate, and its range and confidence.
- Offer real options: reduce scope, phase delivery, add resources, or accept the risk explicitly.
- The sponsor chooses between real trade-offs. The estimate stays honest.

> ON THE EXAM
>
> Negotiate scope, time, resources or risk. Never negotiate the estimate itself.

Project Lantern: estimating two ways

- **Kiosk installation**: parametric, at a supplier rate per kiosk, checked against the first clinic's actuals.
- **Data migration**: three-point, because data quality was uncertain. PERT 18 days, standard deviation 3.
- **Booking app**: story points and planning poker; release forecast from measured velocity.
- Every estimate recorded with its basis and range, and revised as actuals arrived.

> ON THE EXAM
>
> The right technique depends on the work and what is known. Hybrid projects use several.

Scenario signals and the move

THE SCENARIO SAYS	THE MOVE
Optimistic, most likely, pessimistic given	PERT: (O + 4M + P) / 6, unless told triangular
Quick early estimate, similar past project	Analogous
Points remaining and recent velocity given	Divide; forecast a range
Sponsor wants the estimate cut	Explain the basis; offer real trade-offs
Planning poker votes far apart	Outliers explain; discuss; re-vote
Estimates keep running over on similar work	Calibrate from history, with the team

ON THE EXAM

Honest ranges, the right technique for what is known, and history used to calibrate.

> A team estimates an activity using three points. Optimistic: 6 days. Most likely: 9 days. Pessimistic: 18 days.
>
> The project uses beta distribution (PERT) estimates.

What is the expected duration of the activity?

A 11 days

B 10 days

C 9 days

D 12 days

◆ PAUSE AND DECIDE

What is the expected duration of the activity?

A 11 days

The triangular average: 33 / 3. The question asked for PERT.

B 10 days

(6 + 4 x 9 + 18) / 6 = 60 / 6 = 10.

C 9 days

The most likely value alone. Ignores the risk of overrun.

D 12 days

The midpoint of optimistic and pessimistic. Not a standard formula here.

ON THE EXAM

PERT: optimistic plus four times most likely plus pessimistic, over six. Check which formula is asked.

> A sponsor needs a rough cost for a new warehouse racking project by tomorrow, for a funding discussion.
>
> The company completed a very similar racking project at another site last year. No detailed design exists yet.

Which estimating technique is most appropriate?

A Analogous estimating, based on last year's similar project.

B Bottom-up estimating of every work package.

C Three-point estimating for every activity.

D A Monte Carlo simulation of the full schedule.

◆ PAUSE AND DECIDE

Which estimating technique is most appropriate?

A Analogous.

Quick, uses a very similar past project, and fits a rough early estimate.

B Bottom-up.

Needs work packages that do not exist yet, and takes far longer than a day.

C Three-point per activity.

Needs a list of activities that does not exist yet.

D Monte Carlo.

Needs a detailed schedule model. Heavy for a rough estimate.

THE SIGNAL

Rough, fast, early, with a similar past project available: analogous.

> A Scrum team has 180 story points left in its release backlog. Its velocity over the last four two-week sprints was 28, 32, 30 and 30.
>
> The product owner asks how long the remaining work is likely to take.

What is the best forecast?

A About 6 weeks

B About 9 weeks

C About 15 weeks

D About 12 weeks

◆ PAUSE AND DECIDE

What is the best forecast?

A About 6 weeks

Six is the number of sprints. The sprints are two weeks each.

B About 9 weeks

Not derived from the data given.

C About 15 weeks

Uses a slower velocity than the team has shown.

D About 12 weeks

Average velocity 30; 180 / 30 = 6 sprints of 2 weeks.

THE SIGNAL

Points divided by average velocity gives sprints. Then convert sprints to weeks.

> A bottom-up estimate for a project comes to $2.4 million, with a documented basis. The sponsor says the budget is $1.7 million and asks the project manager to 'find a way to make the estimate fit'.
>
> The scope has not changed.

What should the project manager do?

A Reduce each work package estimate proportionally so the total is $1.7 million.

B Submit the $1.7 million figure, and hold the difference as hidden contingency.

C Explain the basis of the estimate, and present options such as reducing scope, phasing, or more funding.

D Decline to discuss the estimate further, since it is already documented.

◆ PAUSE AND DECIDE

What should the project manager do?

A Scale every estimate down.

The work has not shrunk. The overrun is merely postponed and hidden.

B Hide the difference.

Dishonest, and the numbers still do not add up.

C Explain the basis; offer real options.

Keeps the estimate honest and gives the sponsor genuine choices.

D Refuse to discuss.

Unhelpful. The sponsor has a real constraint that needs solving together.

THE SIGNAL

Same scope, smaller number requested: negotiate scope, time, funding or risk, never the estimate.

> During planning poker, most developers show 5 for a story. One shows 2, and one shows 13.
>
> The team is keen to move on to the next story.

What should happen next?

A The developers who showed 2 and 13 explain their reasoning, the team discusses, then votes again.

B Take the average of all the votes and move on.

C Use the highest estimate, 13, to be safe.

D The project leader chooses the estimate to save time.

◆ PAUSE AND DECIDE

What should happen next?

A Outliers explain; discuss; re-vote.

Surfaces the risk or shortcut behind the outliers. That is the point of the technique.

B Average the votes.

Throws away the information in the outliers.

C Take the highest.

Pads without understanding. The 13 might be based on a misunderstanding.

D The leader decides.

Estimates belong to the developers doing the work.

THE SIGNAL

Divergent votes carry information. Let the outliers explain before converging.

> Over the last three projects, a team's estimates for data integration work ran about forty percent below the actual effort each time.
>
> The team is now estimating integration work for a new project.

What should the project manager do?

A. Add forty percent to every estimate the team gives.

B. Review the past actuals with the team, understand the causes, and calibrate the new estimates.

C. Tell the team its estimates are unreliable and must improve.

D. Stop estimating integration work and schedule it as fixed blocks of time.

◆ PAUSE AND DECIDE

What should the project manager do?

A. Add forty percent everywhere.
Blanket padding without understanding. The cause may apply to only some work.

B. Review actuals with the team; calibrate.
Uses historical data, finds the cause, and improves estimates on evidence.

C. Tell the team to improve.
Blame without analysis. Nothing changes.

D. Stop estimating.
Abandons planning information because it was inaccurate, instead of improving it.

THE SIGNAL

A consistent pattern in history is a gift. Find its cause with the team, then calibrate.

Honest ranges. The right technique for what is known. History to calibrate.

admin@sthithapragna.com